

- Description

This module manages a smart refrigerator defrost cycle optimization system, ensuring that:

1. The system continuously monitors the frost thickness (in mm) on the evaporator.
2. If the frost thickness exceeds a predefined threshold (e.g., 5 mm), the defrost cycle is automatically activated.
3. If the frost thickness is below the threshold, no defrost action is take.

This system helps maintain optimal cooling efficiency and energy consumption by preventing excessive frost buildup.

- Specification

module SmartRefrigeratorDefrostControl

var currentFrostThickness, defrostThreshold: real

var defrostCycleActive: boolean

process updateFrostThickness(newThickness: real) returns (currentFrostThickness: real, defrostCycleActive: boolean)

functional scenario form:

 currentFrostThickness > defrostThreshold && defrostCycleActive = true,
output "Frost thickness high: " + currentFrostThickness + " mm. Defrost cycle activated.",
Defrost cycle activated.

 ||

 currentFrostThickness <= defrostThreshold && defrostCycleActive = false,
output "Frost thickness normal: " + currentFrostThickness + " mm. No defrost needed.",
No defrost cycle triggered.

end process

end module

- Java Code

```
package code;
public class c88_SmartRefrigerator {
    private double defrostThreshold;
    private double currentFrostThickness;
    private boolean defrostCycleActive;
    public c88_SmartRefrigerator(double defrostThreshold, double initialFrostThickness) {
        this.defrostThreshold = defrostThreshold;
        this.currentFrostThickness = initialFrostThickness;
        this.defrostCycleActive = false;
    }
    public void updateFrostThickness(double newFrostThickness) {
        currentFrostThickness = newFrostThickness;
        adjustDefrostCycle();
    }
}
```

```

        System.out.printf("Current Frost Thickness: %.2f mm, Defrost Cycle: %s\n",
                           currentFrostThickness, defrostCycleActive ? "ACTIVE" :
                           "INACTIVE");
        if (defrostCycleActive) {
            System.out.println("ALERT: Frost thickness exceeds threshold. Defrost cycle activated.");
        } else {
            System.out.println("Frost thickness is within acceptable limits. No defrost needed.");
        }
    }
    private void adjustDefrostCycle() {
        if (currentFrostThickness > defrostThreshold) {
            defrostCycleActive = true;
        } else {
            defrostCycleActive = false;
        }
    }
    public double getCurrentFrostThickness() {
        return currentFrostThickness;
    }
    public boolean isDefrostCycleActive() {
        return defrostCycleActive;
    }
    public static void main(String[] args) {
        c88_SmartRefrigerator fridge = new c88_SmartRefrigerator(5.0, 3.0)
;
        // testcase-VT:
        //     fridge.updateFrostThickness(4.0);
        //     fridge.updateFrostThickness(6.5);
        //     fridge.updateFrostThickness(5.0);

        // testcase-FT:
        //     fridge.updateFrostThickness(-1.7976931344437577E308);
        //     fridge.updateFrostThickness(-2.2250738585072014E-308);
        //     fridge.updateFrostThickness(1.3834084758073603E308);

        // testcase-Z3:
        //     fridge.updateFrostThickness(4.0);
        //     fridge.updateFrostThickness(6.5);
        //     fridge.updateFrostThickness(5.0);

        // testcase-UVT:

```

```

        fridge.updateFrostThickness(4.0);
        fridge.updateFrostThickness(6.5);
        fridge.updateFrostThickness(5.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c88_SmartRefrigerator.java	100.0 %	6	0	6
▼ c88_SmartRefrigerator	100.0 %	6	0	6
main(String[])		0	0	0
c88_SmartRefrigerator(double, c		0	0	0
adjustDefrostCycle()	100.0 %	2	0	2
updateFrostThickness(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 92.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	79	21,492	21,571
▼ src	0.4 %	79	21,492	21,571
▼ code	0.4 %	79	21,492	21,571
▼ c88_SmartRefrigerator.java	92.9 %	79	6	85
▼ c88_SmartRefrigerator	92.9 %	79	6	85
main(String[])	100.0 %	16	0	16
c88_SmartRefrigerator(double, c	100.0 %	12	0	12
adjustDefrostCycle()	100.0 %	14	0	14
updateFrostThickness(double)	100.0 %	37	0	37

4. Fuzz testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c88_SmartRefrigerator.java	100.0 %	6	0	6
▼ c88_SmartRefrigerator	100.0 %	6	0	6
main(String[])		0	0	0
c88_SmartRefrigerator(double, c		0	0	0
adjustDefrostCycle()	100.0 %	2	0	2
updateFrostThickness(double)	100.0 %	4	0	4

5. Fuzz testing instruction coverage: 92.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	79	21,492	21,571
▼ src	0.4 %	79	21,492	21,571
▼ code	0.4 %	79	21,492	21,571
▼ c88_SmartRefrigerator.java	92.9 %	79	6	85
▼ c88_SmartRefrigerator	92.9 %	79	6	85
main(String[])	100.0 %	16	0	16
c88_SmartRefrigerator(double, c	100.0 %	12	0	12
adjustDefrostCycle()	100.0 %	14	0	14
updateFrostThickness(double)	100.0 %	37	0	37

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c88_SmartRefrigerator.java	100.0 %	6	0	6
▼ c88_SmartRefrigerator	100.0 %	6	0	6
main(String[])		0	0	0
c88_SmartRefrigerator(double, c		0	0	0
adjustDefrostCycle()	100.0 %	2	0	2
updateFrostThickness(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 92.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	79	21,492	21,571
▼ src	0.4 %	79	21,492	21,571
▼ code	0.4 %	79	21,492	21,571
▼ c88_SmartRefrigerator.java	92.9 %	79	6	85
▼ c88_SmartRefrigerator	92.9 %	79	6	85
main(String[])	100.0 %	16	0	16
c88_SmartRefrigerator(double, c	100.0 %	12	0	12
adjustDefrostCycle()	100.0 %	14	0	14
updateFrostThickness(double)	100.0 %	37	0	37

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c88_SmartRefrigerator.java	100.0 %	6	0	6
▼ c88_SmartRefrigerator	100.0 %	6	0	6
main(String[])		0	0	0
c88_SmartRefrigerator(double, c		0	0	0
adjustDefrostCycle()	100.0 %	2	0	2
updateFrostThickness(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 92.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	79	21,492	21,571
▼ src	0.4 %	79	21,492	21,571
▼ code	0.4 %	79	21,492	21,571
▼ c88_SmartRefrigerator.java	92.9 %	79	6	85
▼ c88_SmartRefrigerator	92.9 %	79	6	85
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c88_SmartRefrigerator(double, c	100.0 %	12	0	12
adjustDefrostCycle()	100.0 %	14	0	14
updateFrostThickness(double)	100.0 %	37	0	37